eBGP, EIGRP and OSPF configuration

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Purpose:

The purpose of this lab is to learn the use of eBGP (external Border Gateway Protocol) and EIGRP (Enhanced Interior Gateway Routing Protocol), both of which are popular routing protocols in the networking world. eBGP being used to unite autonomous systems, aka company networks, and EIGRP which is used for in-network routing, combined these are a potent team in modern networking to tackle complex problems with limited networking overhead.

Background Information:

Border Gateway Protocol (BGP) is an essential routing protocol used for transferring data across the internet, specifically between large networks, known as autonomous systems (ASs), such as major enterprise networks. BGP plays a critical role in managing the complex paths that data takes as it travels across these networks, ensuring that data packets reach their destination efficiently. It has two main modes: internal BGP (iBGP) for communication within a single AS, and external BGP (eBGP) for routing data between different ASs. BGP relies on TCP (Transmission Control Protocol) over port 179, which makes it reliable because TCP itself ensures data packets are properly delivered and in the right order. One of the primary features of eBGP is its path-vector mechanism, which maintains a record of AS numbers, allowing it to avoid routing loops by tracking where data has been. This feature also makes it highly scalable for complex networks, as it can handle vast amounts of routing information. eBGP is known for its policy-based routing, which allows network administrators to set up rules that dictate data flow based on factors beyond simple distance, such as security policies or business agreements. This scalability and control make eBGP essential to the internet's infrastructure.

Enhanced Interior Gateway Routing Protocol (EIGRP) is a Cisco-proprietary protocol, meaning it was originally designed only for Cisco devices, although it's

now sometimes used more widely. EIGRP combines elements from both distance-vector and link-state protocols, and it's designed specifically for routing within a single AS. EIGRP is widely used in large enterprise networks because of its speed and efficiency. Its Diffusing Update Algorithm (DUAL) ensures that there are loop-free paths and that the network converges quickly, which means it can adapt to changes, like new routes or failed links, in real-time without causing delays. Instead of sending out entire routing tables like some other protocols, EIGRP only sends updates when there's a change, which conserves bandwidth and reduces unnecessary network traffic. EIGRP also uses a unique composite metric, which takes into account factors like bandwidth, delay, load, and reliability, rather than just distance. This makes it easier for network administrators to find the most efficient path for data while also managing the reliability and load of each route.

Open Shortest Path First (OSPF) is an open-standard, link-state protocol that's commonly used for routing within ASs, especially in multi-vendor environments, meaning networks that don't rely on just one manufacturer's devices. OSPF creates a complete map of the network using a synchronized link-state database (LSDB) that all OSPF routers share by exchanging link-state advertisements (LSAs). This topological map of the network helps each router have a full view of the entire network structure, allowing them to calculate optimal routes based on the entire network's layout. OSPF is also highly organized and scalable, thanks to its hierarchical design. It uses areas to group routers and links, making it possible to reduce the load and traffic within large networks by focusing updates in specific parts of the network rather than everywhere at once, however we are using only one area of OSPF. For determining the shortest path, OSPF relies on Dijkstra's Shortest Path First (SPF) algorithm, which finds the least-cost route to each destination. The cost is typically determined by bandwidth, but it can be adjusted to consider other factors if needed.

Overall, eBGP, EIGRP, and OSPF are all powerful protocols that serve different purposes and work to create potent combination in modern networking. eBGP is used for inter-AS routing between large networks on the internet because of its policy controls and scalability. EIGRP, meanwhile, is designed for efficient and fast routing within Cisco-based networks, making it a go-to for many enterprises. Finally, OSPF provides reliable, organized, and flexible routing within ASs and is especially popular in multi-vendor networks. Knowing how these protocols work and when to use each one is important for configuring and maintaining complex networks that need to handle large amounts of traffic efficiently.

Summary:

In this lab I configure 6 routers, 2 separate EIGRP instances, 2 separate BGP instances, and 1 OSPF instance, this is meant to model the connection of corporate networks, being connected by the BGP instances for interconnectivity between the networks. The BGP took and gave routes to both the EIGRP and OSPF instances to allow end to end connectivity.

Commands:

Routers 1 & 6

hostname R1 – assigns hostname to router
ipv6 unicast-routing – allows ipv6 to be routed
interface GigabitEthernet0/0/0 – allows you to configure the interface
ip address 192.168.1.1 255.255.255.252 – assigns ip address based on address table
ipv6 address FE80::1 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:B::1/64 – assigns ipv6 address based on address table
ipv6 enable – allows the interface to run ipv6 routing protocols
ipv6 eigrp 10 – configures the interface with ipv6 eigrp
interface GigabitEthernet0/0/1 – allows you to configure the interface
ip address 192.168.0.2 255.255.255.0 – assigns ip address based on address table
ipv6 address FE80::2 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:A::2/64 – assigns interface ipv6 address based on address table
ipv6 enable – enables interface to run ipv6 routing
ipv6 eigrp 10 – configures ipv6 eigrp on the interface
router eigrp 10 – allows configuration of the eigrp routing protocol
network 192.168.0.0 0.0.0.3 – allows the network stated to run eigrp through the router
network 192.168.1.0 0.0.0.3 – allows the network stated to run eigrp through the router
eigrp router-id 1.1.1.1 – assigns the eigrp router-id
ipv6 router eigrp 10 – allows configuration of the eigrp routing protocol
eigrp router-id 1.1.1.1 – assigns the ipv6 router-id

Routers 2 & 5

hostname R2 – assigns hostname to router
ipv6 unicast-routing – allows ipv6 to be routed
interface GigabitEthernet0/0/0 – allows you to configure the interface
ip address 192.168.2.1 255.255.255.252 – assigns ip address based on address table
ipv6 address FE80::2 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:C::1/64 – assigns ipv6 address based on address table
ipv6 enable – enables ipv6 routing protocols on the interface
ipv6 eigrp 10 – sets up ipv6 eigrp on this interface
interface GigabitEthernet0/0/1 – allows you to configure the interface
ip address 192.168.1.2 255.255.255.252 – assigns ip address based on address table
ipv6 address FE80::2 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:B::2/64 – assigns ipv6 address based on address table
ipv6 enable – enables ipv6 routing protocols on the interface
ipv6 eigrp 10 – configures ipv6 eigrp to run on the interface
router eigrp 10 – allows configuration of eigrp with autonomous system number 10
network 192.168.1.0 0.0.0.3 – allows network stated to run eigrp through the router
network 192.168.2.0 0.0.0.3 – allows network stated to run eigrp through the router
redistribute connected metric 10 0 255 255 1 – redistributes the networks on router to eBGP
redistribute BGP 2 metric 10 0 255 255 1 – redistributes eBGP networks back to eigrp network

eigrp router-id 2.2.2.2 – sets router id for eigrp
router BGP 2 – allows configuration of BGP
BGP router-id 2.2.2.2 – sets the BGP router id
BGP log-neighbor-changes – sends notification and logs when a neighbor adjacency is set up or broken
neighbor 2001:DB8:ACAD:C::2 remote-as 3 – declares set ip address as the neighbor
neighbor 192.168.2.2 remote-as 3 – sets up to look for the ip address of neighbor with matching ip's
address-family ipv4 – allows specific configuration of ipv4 BGP
redistribute connected metric 10 – sets connects redistribution of AS of the router and BGP
redistribute eigrp 10 metric 10 – redistributes the eigrp network to BGP
no neighbor 2001:DB8:ACAD:C::2 activate – makes sure ipv6 neighbor doesn't overlap ipv4
neighbor 192.168.2.2 activate – allows ipv4 neighbor to share and shares routes with that neighbor
exit-address-family – exits address family
address-family ipv6 – allows configuration of ipv6 BGP specifically
redistribute connected metric 10 – connects redistribution of AS and BGP
redistribute eigrp 10 metric 10 – redistributes the eigrp networks to BGP
neighbor 2001:DB8:ACAD:C::2 activate – allows the ipv6 neighbor to add routes
exit-address-family – exits address family
ipv6 router eigrp 10 – allows configuration of ipv6 eigrp
eigrp router-id 2.2.2.2 – sets the router id of ipv6 eigrp
redistribute BGP 2 metric 10 0 255 255 1 – allows redistribution of BGP routes to the eigrp networks

Routers 3 & 4

hostname R3 – assigns hostname to router
ipv6 unicast-routing – allows ipv6 to be routed
interface GigabitEthernet0/0/0 – allows you to configure the interface
ip address 192.168.3.1 255.255.255.252 – assigns ip address based on address table
ip OSPF 10 area 0 – configures the interface with OSPF and the area
ipv6 address FE80::1 link-local – assigns ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:D::1/64 – assigns ipv6 address based on address table
ipv6 enable – enable ipv6 configuration on interface
ipv6 OSPF 10 area 0 – configures ipv6 OSPF and area
interface GigabitEthernet0/0/1 – allows you to configure the interface
ip address 192.168.2.2 255.255.255.252 – configures ip address based on address table
ipv6 address FE80::1 link-local – configures ipv6 link-local address based on address table
ipv6 address 2001:DB8:ACAD:C::2/64 – configures ipv6 address based on address table
router OSPF 10 – allows configuration of OSPF
router-id 3.3.3.3 – configures opsf router id
redistribute BGP 3 metric 10 subnets – redistributes the BGP routes to OSPF
network 192.168.2.0 0.0.0.3 area 0 – sets up OSPF networks to neighboring routers
network 192.168.3.0 0.0.0.3 area 0 – sets up OSPF networks to neighboring routers
router BGP 3 – allows configuration to BGP
BGP router-id 3.3.3.3 – configures BGP router id
BGP log-neighbor-changes – logs and notifies of neighbor adjacency set ups and break downs
neighbor 2001:DB8:ACAD:C::1 remote-as 2 – sets ipv6 neighbor
neighbor 192.168.2.1 remote-as 2 – sets ipv4 neighbor
address-family ipv4 – allows specific ipv4 configurations to BGP
redistribute connected – connects redistributions from OSPF and BGP
redistribute OSPF 10 metric 10 – redistributes OSPF routes to BGP
redistribute OSPF 10 metric 10 match external 2 – redistributes OSPF routes to BGP
redistribute OSPF 10 metric 10 match internal – redistributes OSPF routes to BGP
no neighbor 2001:DB8:ACAD:C::1 activate – makes ipv6 neighbor not interfere with ipv4
neighbor 192.168.2.1 activate – sets ipv4 neighbor communication
exit-address-family – exits address family
address-family ipv6 – allows specific ipv4 configurations to BGP
redistribute connected – connects redistribution from OSPF and BGP
redistribute OSPF 10 metric 10 – redistributes OSPF routes to BGP
redistribute OSPF 10 metric 10 match external 2 – redistributes OSPF routes to BGP
redistribute OSPF 10 metric 10 match internal – redistributes OSPF routes to BGP
neighbor 2001:DB8:ACAD:C::1 activate – activates ipv6 neighbor
exit address-family – exits address family
ipv6 router OSPF 10 – allows ipv6 configurations for OSPF
router-id 3.3.3.3 – sets opsfv3 router id
redistribute BGP 3 – redistributes ipv6 BGP routes

Configurations:

Router1:

```
Current configuration : 1713 bytes
Last configuration change at 23:22:27 UTC Thu Nov 7 2024
```

```
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R1
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM2408005M
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.1.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:B::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
ip address 192.168.0.2 255.255.255.0
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:A::2/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
```

```
negotiation auto
router eigrp 10
network 192.168.0.0
network 192.168.1.0 0.0.0.3
eigrp router-id 1.1.1.1
ip forward-protocol nd
no ip http server
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 10
eigrp router-id 1.1.1.1
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router2:

```
Current configuration : 2291 bytes
Last configuration change at 23:06:56 UTC Thu Nov 7 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R2
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
```

```
mode none
interface GigabitEthernet0/0/0
ip address 192.168.2.1 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:C::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
ip address 192.168.1.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:B::2/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/2/0
no ip address
negotiation auto
interface GigabitEthernet0/2/1
no ip address
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
router eigrp 10
network 192.168.1.0 0.0.0.3
network 192.168.2.0 0.0.0.3
redistribute connected metric 10 0 255 255 1
redistribute BGP 2 metric 10 0 255 255 1
eigrp router-id 2.2.2.2
router BGP 2
BGP router-id 2.2.2.2
BGP log-neighbor-changes
neighbor 2001:DB8:ACAD:C::2 remote-as 3
neighbor 192.168.2.2 remote-as 3
address-family ipv4
redistribute connected metric 10
redistribute eigrp 10 metric 10
no neighbor 2001:DB8:ACAD:C::2 activate
neighbor 192.168.2.2 activate
exit-address-family
address-family ipv6
redistribute connected metric 10
redistribute eigrp 10 metric 10
neighbor 2001:DB8:ACAD:C::2 activate
exit-address-family
```

```
ip forward-protocol nd
no ip http server
ip http secure-server
ipv6 router eigrp 10
eigrp router-id 2.2.2.2
redistribute BGP 2 metric 10 0 255 255 1
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router3:

```
Current configuration : 2109 bytes
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R3
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214420G7
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
ip address 192.168.3.1 255.255.255.252
ip OSPF 10 area 0
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:D::1/64
ipv6 enable
ipv6 OSPF 10 area 0
interface GigabitEthernet0/0/1
```

```
ip address 192.168.2.2 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:C::2/64
interface Serial0/1/0
interface Serial0/1/1
```

```
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
interface Vlan1
no ip address
shutdown
router OSPF 10
router-id 3.3.3.3
redistribute BGP 3 metric 10 subnets
network 192.168.2.0 0.0.0.3 area 0
network 192.168.3.0 0.0.0.3 area 0
router BGP 3
BGP router-id 3.3.3.3
BGP log-neighbor-changes
neighbor 2001:DB8:ACAD:C::1 remote-as 2
neighbor 192.168.2.1 remote-as 2
address-family ipv4
redistribute connected
redistribute OSPF 10 metric 10 match internal external 2
no neighbor 2001:DB8:ACAD:C::1 activate
neighbor 192.168.2.1 activate
exit-address-family
address-family ipv6
redistribute connected
redistribute OSPF 10 metric 10 match internal external 2
neighbor 2001:DB8:ACAD:C::1 activate
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router OSPF 10
router-id 3.3.3.3
redistribute BGP 3
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router4:

```
Current configuration : 2180 bytes
Last configuration change at 22:48:35 UTC Thu Nov 7 2024
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R4
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO211216BL
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
ip address 192.168.4.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:E::1/64
interface GigabitEthernet0/0/1
ip address 192.168.3.2 255.255.255.252
ip OSPF 10 area 0
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:D::2/64
ipv6 enable
ipv6 OSPF 10 area 0
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
interface Vlan1
```

```
no ip address
router OSPF 10
router-id 4.4.4.4
redistribute BGP 4 metric 10 subnets
network 192.168.3.0 0.0.0.3 area 0
network 192.168.4.0 0.0.0.3 area 0
router BGP 4
BGP router-id 4.4.4.4
BGP log-neighbor-changes
neighbor 2001:DB8:ACAD:E::2 remote-as 5
neighbor 192.168.4.2 remote-as 5
address-family ipv4
redistribute connected
redistribute OSPF 10 metric 10 match internal external 2
no neighbor 2001:DB8:ACAD:E::2 activate
neighbor 192.168.4.2 activate
exit-address-family
address-family ipv6
redistribute connected
redistribute OSPF 10 metric 10 match internal external 2
neighbor 2001:DB8:ACAD:E::2 activate
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router OSPF 10
router-id 4.4.4.4
redistribute BGP 4
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Router5:

```
Current configuration : 2227 bytes
Last configuration change at 22:47:07 UTC Thu Nov 7 2024
version 15.5
service timestamps debug datetime msec
service timestamps log datetime msec
no platform punt-keepalive disable-kernel-core
hostname R5
```

```
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
ipv6 unicast-routing
subscriber templating
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FDO214421CF
spanning-tree extend system-id
redundancy
mode none
vlan internal allocation policy ascending
interface GigabitEthernet0/0/0
ip address 192.168.5.1 255.255.255.252
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:F::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
ip address 192.168.4.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:E::2/64
ipv6 enable
ipv6 eigrp 10
interface Serial0/1/0
no ip address
interface Serial0/1/1
no ip address
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
negotiation auto
interface Vlan1
no ip address
router eigrp 10
network 192.168.4.0 0.0.0.3
network 192.168.5.0 0.0.0.3
redistribute connected metric 10 0 255 255 1
redistribute BGP 5 metric 10 0 255 255 1
eigrp router-id 5.5.5.5
router BGP 5
```

```
BGP router-id 5.5.5.5
BGP log-neighbor-changes
neighbor 2001:DB8:ACAD:E::1 remote-as 4
neighbor 192.168.4.1 remote-as 4
address-family ipv4
redistribute connected metric 10
redistribute eigrp 10 metric 10
no neighbor 2001:DB8:ACAD:E::1 activate
neighbor 192.168.4.1 activate
exit-address-family
address-family ipv6
redistribute connected metric 10
redistribute eigrp 10 metric 10
neighbor 2001:DB8:ACAD:E::1 activate
exit-address-family
ip forward-protocol nd
no ip http server
no ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 10
eigrp router-id 5.5.5.5
redistribute BGP 5 metric 10 0 255 255 1
control-plane
line con 0
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

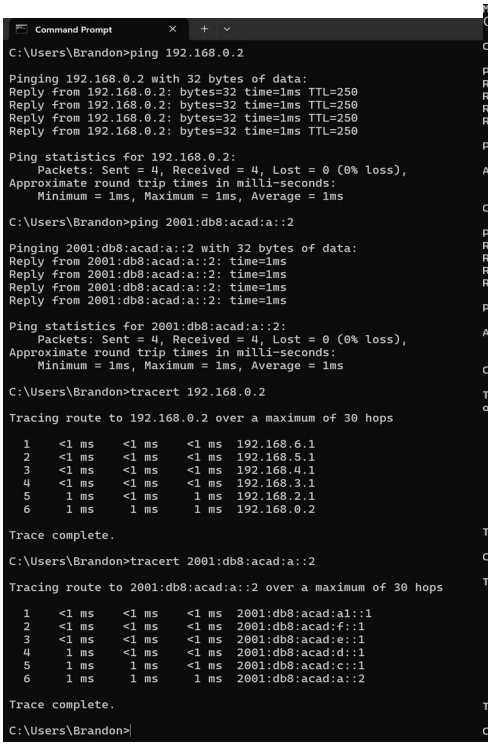
Router6:

```
Current configuration : 1713 bytes
Last configuration change at 23:26:32 UTC Thu Nov 7 2024
version 16.9
service timestamps debug datetime msec
service timestamps log datetime msec
platform qfp utilization monitor load 80
platform punt-keepalive disable-kernel-core
hostname R6
boot-start-marker
boot-end-marker
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
```

```
vrf definition Mgmt-intf
address-family ipv4
exit-address-family
address-family ipv6
exit-address-family
no aaa new-model
login on-success log
subscriber templating
ipv6 unicast-routing
multilink bundle-name authenticated
license udi pid ISR4321/K9 sn FLM24060912
no license smart enable
diagnostic bootup level minimal
spanning-tree extend system-id
redundancy
mode none
interface GigabitEthernet0/0/0
ip address 192.168.6.1 255.255.255.0
negotiation auto
ipv6 address FE80::1 link-local
ipv6 address 2001:DB8:ACAD:A1::1/64
ipv6 enable
ipv6 eigrp 10
interface GigabitEthernet0/0/1
ip address 192.168.5.2 255.255.255.252
negotiation auto
ipv6 address FE80::2 link-local
ipv6 address 2001:DB8:ACAD:F::2/64
ipv6 enable
ipv6 eigrp 10
interface Serial0/1/0
interface Serial0/1/1
interface GigabitEthernet0/2/0
no ip address
shutdown
negotiation auto
interface GigabitEthernet0/2/1
no ip address
shutdown
negotiation auto
interface GigabitEthernet0
vrf forwarding Mgmt-intf
no ip address
shutdown
negotiation auto
router eigrp 10
network 192.168.5.0
```

```
network 192.168.6.0
eigrp router-id 6.6.6.6
ip forward-protocol nd
ip http server
ip http authentication local
ip http secure-server
ip tftp source-interface GigabitEthernet0
ipv6 router eigrp 10
eigrp router-id 6.6.6.6
control-plane
line con 0
transport input none
stopbits 1
line aux 0
stopbits 1
line vty 0 4
login
end
```

Screenshots:



```
Command Prompt

C:\Users\Brandon>ping 192.168.0.2

Pinging 192.168.0.2 with 32 bytes of data:
Reply from 192.168.0.2: bytes=32 time=1ms TTL=250
Reply from 192.168.0.2: bytes=32 time=1ms TTL=250
Reply from 192.168.0.2: bytes=32 time=1ms TTL=250
Reply from 192.168.0.2: bytes=32 time=1ms TTL=250

Ping statistics for 192.168.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Brandon>ping 2001:db8:acad:a::2

Pinging 2001:db8:acad:a::2 with 32 bytes of data:
Reply from 2001:db8:acad:a::2: time=1ms
Reply from 2001:db8:acad:a::2: time=1ms
Reply from 2001:db8:acad:a::2: time=1ms
Reply from 2001:db8:acad:a::2: time=1ms

Ping statistics for 2001:db8:acad:a::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Brandon>tracert 192.168.0.2

Tracing route to 192.168.0.2 over a maximum of 30 hops

  1    <1 ms    <1 ms    <1 ms  192.168.6.1
  2    <1 ms    <1 ms    <1 ms  192.168.5.1
  3    <1 ms    <1 ms    <1 ms  192.168.4.1
  4    <1 ms    <1 ms    <1 ms  192.168.3.1
  5     1 ms    <1 ms     1 ms  192.168.2.1
  6     1 ms     1 ms     1 ms  192.168.0.2

Trace complete.

C:\Users\Brandon>tracert 2001:db8:acad:a::2

Tracing route to 2001:db8:acad:a::2 over a maximum of 30 hops

  1    <1 ms    <1 ms    <1 ms  2001:db8:acad:a1::1
  2    <1 ms    <1 ms    <1 ms  2001:db8:acad:f::1
  3    <1 ms    <1 ms    <1 ms  2001:db8:acad:e::1
  4     1 ms    <1 ms    <1 ms  2001:db8:acad:d::1
  5     1 ms     1 ms    <1 ms  2001:db8:acad:c::1
  6     1 ms     1 ms     1 ms  2001:db8:acad:a::2

Trace complete.

C:\Users\Brandon>
```

```
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Alex Huhdanpaa>ping 192.168.6.2

Pinging 192.168.6.2 with 32 bytes of data:
Reply from 192.168.6.2: bytes=32 time=1ms TTL=122
Reply from 192.168.6.2: bytes=32 time=1ms TTL=122
Reply from 192.168.6.2: bytes=32 time=1ms TTL=122
Reply from 192.168.6.2: bytes=32 time=1ms TTL=122

Ping statistics for 192.168.6.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Alex Huhdanpaa>ping 2001:db8:acad:a1::2

Pinging 2001:db8:acad:a1::2 with 32 bytes of data:
Reply from 2001:db8:acad:a1::2: time=1ms
Reply from 2001:db8:acad:a1::2: time=1ms
Reply from 2001:db8:acad:a1::2: time=1ms
Reply from 2001:db8:acad:a1::2: time=1ms

Ping statistics for 2001:db8:acad:a1::2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Users\Alex Huhdanpaa>tracert 192.168.6.2

Tracing route to DESKTOP-JT6LALA [192.168.6.2]
over a maximum of 30 hops:

  1    <1 ms    <1 ms    <1 ms  192.168.0.2
  2    <1 ms    <1 ms    <1 ms  192.168.1.2
  3    <1 ms    <1 ms    <1 ms  192.168.2.2
  4    <1 ms    <1 ms    <1 ms  192.168.3.2
  5    <1 ms    <1 ms    <1 ms  192.168.4.2
  6     1 ms     1 ms     1 ms  192.168.5.2
  7     1 ms     1 ms    <1 ms  DESKTOP-JT6LALA [192.168.6.2]

Trace complete.

C:\Users\Alex Huhdanpaa>tracert 2001:db8:acad:a1::2

Tracing route to 2001:db8:acad:a1::2 over a maximum of 30 hops

  1     1 ms    <1 ms    <1 ms  2001:db8:acad:a::2
  2     1 ms    <1 ms    <1 ms  2001:db8:acad:b::2
  3     1 ms    <1 ms    <1 ms  2001:db8:acad:c::2
  4     1 ms    <1 ms    <1 ms  2001:db8:acad:d::2
  5     1 ms     1 ms    <1 ms  2001:db8:acad:e::2
  6     1 ms     1 ms     1 ms  2001:db8:acad:f::2
  7     1 ms     1 ms     1 ms  2001:db8:acad:a1::2

Trace complete.

C:\Users\Alex Huhdanpaa>
```

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

      192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.0.0/24 is directly connected, GigabitEthernet0/0/1
L        192.168.0.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.1.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.2.0/30 is subnetted, 1 subnets
D        192.168.2.0 [90/3072] via 192.168.1.2, 00:18:09, GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
D EX     192.168.3.0
           [170/256000256] via 192.168.1.2, 00:18:02, GigabitEthernet0/0/0
      192.168.4.0/30 is subnetted, 1 subnets
D EX     192.168.4.0
           [170/256000256] via 192.168.1.2, 00:11:25, GigabitEthernet0/0/0
      192.168.5.0/30 is subnetted, 1 subnets
D EX     192.168.5.0
           [170/256000256] via 192.168.1.2, 00:17:02, GigabitEthernet0/0/0
D EX  192.168.6.0/24
           [170/256000256] via 192.168.1.2, 00:17:02, GigabitEthernet0/0/0
R1#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
C   2001:DB8:ACAD:A::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:A::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:B::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:B::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
D   2001:DB8:ACAD:C::/64 [90/3072]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:D::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:F::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
EX  2001:DB8:ACAD:A1::/64 [170/256000256]
     via FE80::2, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R1#
```

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D     192.168.0.0/24 [90/3072] via 192.168.1.1, 00:19:52, GigabitEthernet0/0/1
      192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.1.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.2.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.3.0/30 is subnetted, 1 subnets
B        192.168.3.0 [20/0] via 192.168.2.2, 00:18:51
      192.168.4.0/30 is subnetted, 1 subnets
B        192.168.4.0 [20/10] via 192.168.2.2, 00:12:14
      192.168.5.0/30 is subnetted, 1 subnets
B        192.168.5.0 [20/10] via 192.168.2.2, 00:17:51
B     192.168.6.0/24 [20/10] via 192.168.2.2, 00:17:51
R2#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
D   2001:DB8:ACAD:A::/64 [90/3072]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:B::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:B::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:C::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:C::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
B   2001:DB8:ACAD:D::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:F::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/0
B   2001:DB8:ACAD:A1::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R2#
```

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

B     192.168.0.0/24 [20/10] via 192.168.2.1, 00:19:19
      192.168.1.0/30 is subnetted, 1 subnets
B        192.168.1.0 [20/10] via 192.168.2.1, 00:19:19
      192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.2.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.2.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.3.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.4.0/30 is subnetted, 1 subnets
O        192.168.4.0 [110/2] via 192.168.3.2, 00:18:42, GigabitEthernet0/0/0
      192.168.5.0/30 is subnetted, 1 subnets
O E2     192.168.5.0 [110/10] via 192.168.3.2, 00:18:42, GigabitEthernet0/0/0
O E2  192.168.6.0/24 [110/10] via 192.168.3.2, 00:18:42, GigabitEthernet0/0/0
R3#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
B   2001:DB8:ACAD:A::/64 [20/10]
     via FE80::2, GigabitEthernet0/0/1
B   2001:DB8:ACAD:B::/64 [20/10]
     via FE80::2, GigabitEthernet0/0/1
C   2001:DB8:ACAD:C::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:C::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:D::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:D::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
OE2 2001:DB8:ACAD:F::/64 [110/1]
     via FE80::2, GigabitEthernet0/0/0
OE2 2001:DB8:ACAD:A1::/64 [110/1]
     via FE80::2, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R3#
```

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

O E2  192.168.0.0/24 [110/10] via 192.168.3.1, 00:19:40, GigabitEthernet0/0/1
      192.168.1.0/30 is subnetted, 1 subnets
O E2     192.168.1.0 [110/10] via 192.168.3.1, 00:19:40, GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 1 subnets
O        192.168.2.0 [110/2] via 192.168.3.1, 00:19:40, GigabitEthernet0/0/1
      192.168.3.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.3.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.3.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.4.1/32 is directly connected, GigabitEthernet0/0/0
      192.168.5.0/30 is subnetted, 1 subnets
B        192.168.5.0 [20/10] via 192.168.4.2, 00:22:16
B     192.168.6.0/24 [20/10] via 192.168.4.2, 00:22:16
R4#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
OE2 2001:DB8:ACAD:A::/64 [110/1]
     via FE80::1, GigabitEthernet0/0/1
OE2 2001:DB8:ACAD:B::/64 [110/1]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:D::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:D::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:E::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:E::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
B   2001:DB8:ACAD:F::/64 [20/10]
     via FE80::2, GigabitEthernet0/0/0
B   2001:DB8:ACAD:A1::/64 [20/10]
     via FE80::2, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R4#
```

```
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

B     192.168.0.0/24 [20/10] via 192.168.4.1, 00:20:16
      192.168.1.0/30 is subnetted, 1 subnets
B        192.168.1.0 [20/10] via 192.168.4.1, 00:20:16
      192.168.2.0/30 is subnetted, 1 subnets
B        192.168.2.0 [20/10] via 192.168.4.1, 00:15:29
      192.168.3.0/30 is subnetted, 1 subnets
B        192.168.3.0 [20/0] via 192.168.4.1, 00:21:01
      192.168.4.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.4.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.4.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/30 is directly connected, GigabitEthernet0/0/0
L        192.168.5.1/32 is directly connected, GigabitEthernet0/0/0
D     192.168.6.0/24 [90/3072] via 192.168.5.2, 00:23:57, GigabitEthernet0/0/0
R5#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
B   2001:DB8:ACAD:A::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:B::/64 [20/10]
     via FE80::1, GigabitEthernet0/0/1
B   2001:DB8:ACAD:D::/64 [20/0]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:E::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:E::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:F::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:F::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
D   2001:DB8:ACAD:A1::/64 [90/3072]
     via FE80::2, GigabitEthernet0/0/0
L   FF00::/8 [0/0]
     via Null0, receive
R5#
```

```
R6#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

D EX  192.168.0.0/24
          [170/256000256] via 192.168.5.1, 00:21:01, GigabitEthernet0/0/1
      192.168.1.0/30 is subnetted, 1 subnets
D EX     192.168.1.0
          [170/256000256] via 192.168.5.1, 00:21:01, GigabitEthernet0/0/1
      192.168.2.0/30 is subnetted, 1 subnets
D EX     192.168.2.0
          [170/256000256] via 192.168.5.1, 00:16:14, GigabitEthernet0/0/1
      192.168.3.0/30 is subnetted, 1 subnets
D EX     192.168.3.0
          [170/256000256] via 192.168.5.1, 00:21:46, GigabitEthernet0/0/1
      192.168.4.0/30 is subnetted, 1 subnets
D        192.168.4.0 [90/3072] via 192.168.5.1, 00:24:46, GigabitEthernet0/0/1
      192.168.5.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.5.0/30 is directly connected, GigabitEthernet0/0/1
L        192.168.5.2/32 is directly connected, GigabitEthernet0/0/1
      192.168.6.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.6.0/24 is directly connected, GigabitEthernet0/0/0
L        192.168.6.1/32 is directly connected, GigabitEthernet0/0/0
R6#show ipv6 route
IPv6 Routing Table - default - 9 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
EX  2001:DB8:ACAD:A::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:B::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
EX  2001:DB8:ACAD:D::/64 [170/256000256]
     via FE80::1, GigabitEthernet0/0/1
D   2001:DB8:ACAD:E::/64 [90/3072]
     via FE80::1, GigabitEthernet0/0/1
C   2001:DB8:ACAD:F::/64 [0/0]
     via GigabitEthernet0/0/1, directly connected
L   2001:DB8:ACAD:F::2/128 [0/0]
     via GigabitEthernet0/0/1, receive
C   2001:DB8:ACAD:A1::/64 [0/0]
     via GigabitEthernet0/0/0, directly connected
L   2001:DB8:ACAD:A1::1/128 [0/0]
     via GigabitEthernet0/0/0, receive
L   FF00::/8 [0/0]
     via Null0, receive
R6#
```

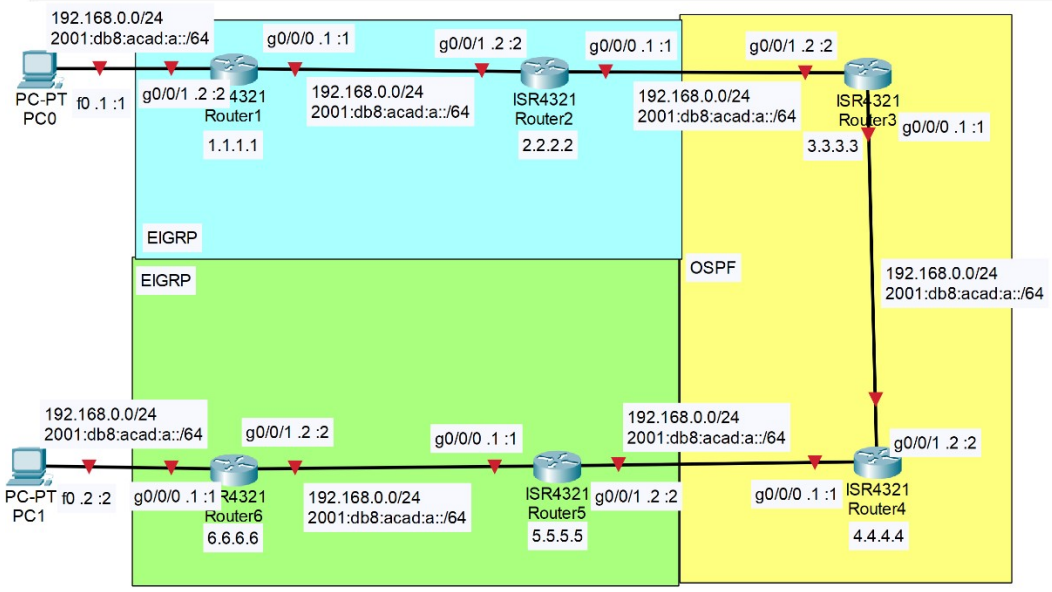
Problems:

The biggest problem I ran in to was routes from BGP not redistributing back in to EIGRP, this was caused by the rigidity and difference between EIGRP and BGP metrics this was resolved by the manual configuration >redistribute connected metric 10 0 255 255 1< this. Another issue I had involved not having any routes to the addresses that were on the same interfaces as BGP addresses, this was resolved as forgetting to use the redistribute OSPF 10 metric 10 match internal command to also redistribute internal OSPF routes and not just external routes to BGP.

Conclusion:

The lab was overall a success albeit with difficulties in the configuration of ipv4 and ipv6 BGP. This was largely due to the fact that I was not aware of BGP

metrics when redistributing routes from BGP to EIGRP or BGP to OSPF, while I was easily able to get routes on to BGP due to the simple nature of it's configurations I was unable to get routes from the BGP table into either of the IGP (Interior Gateway Protocol) until I discovered the necessary metrics to put after the redistribute command.

Diagram and Addressing:

Device name	IP protocol	Interface	IPv6 Address	IPv4 Address	Subnet mask
PC1	----	F0	2001:db8:acad:a::1/64	192.168.0.1	255.255.255.0
R1	----	G0/0/1	2001:db8:acad:a::2/64	192.168.0.2	255.255.255.0
R1	EIGRP	G0/0/0	2001:db8:acad:b::1/64	192.168.1.1	255.255.255.252
R2	EIGRP	G0/0/1	2001:db8:acad:b::2/64	192.168.1.2	255.255.255.252
R2	EIGRP/BGP	G0/0/0	2001:db8:acad:c::1/64	192.168.2.1	255.255.255.252
R3	OSPF/BGP	G0/0/1	2001:db8:acad:c::2/64	192.168.2.2	255.255.255.252
R3	OSPF	G0/0/0	2001:db8:acad:d::1/64	192.168.3.1	255.255.255.252
R4	OSPF	G0/0/1	2001:db8:acad:d::2/64	192.168.3.2	255.255.255.252
R4	OSPF/BGP	G0/0/0	2001:db8:acad:e::1/64	192.168.4.1	255.255.255.252
R5	EIGRP/BGP	G0/0/1	2001:db8:acad:e::2/64	192.168.4.2	255.255.255.252
R5	EIGRP	G0/0/0	2001:db8:acad:f::1/64	192.168.5.1	255.255.255.252
R6	EIGRP	G0/0/1	2001:db8:acad:f::2/64	192.168.5.2	255.255.255.252
R6	----	G0/0/0	2001:db8:acad:a1::1/64	192.168.6.1	255.255.255.0
PC2	----	F0	2001:db8:acad:a1::2/64	192.168.6.2	255.255.255.0